

Question 4

$$1. \quad \hat{y} = b \cdot x = b_0 + \sum_{k=1}^K b_k x_k$$

$$\frac{\partial \hat{y}}{\partial x_k} = b_k$$

For a 1 unit change, the prediction changes by b_k .

2.

$$\hat{p}_i = \frac{e^{bx}}{1 + e^{bx}}$$

$$\frac{\partial}{\partial L} \frac{e^L}{1 + e^L} = \frac{e^L}{1 + e^L} \left(1 - \frac{e^L}{1 + e^L} \right)$$

→ using chain rule

$$\frac{\partial \hat{p}_i}{\partial x_i} = \frac{e^{bx}}{1 + e^{bx}} \left(1 - \frac{e^{bx}}{1 + e^{bx}} \right) b$$

$$\frac{\partial \hat{p}_i}{\partial x_i} = \hat{p}(1 - \hat{p})b$$

→ This does change as x changes

It is the answer from 1 multiplied by $\hat{p}(1 - \hat{p})$
x-dependent

$$\text{where is } 4[\hat{p}(1 - \hat{p})b] \approx b \quad ?$$

$$@ \quad \hat{p} = 0.5$$

$$4[0.5(1 - 0.5)b] = b$$

$$b = b$$

When $\hat{p}$ is around or equal to 0.5, 4 times the coefficient is a good estimate.

3.

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = bx_k$$

log odds ratio

A 1 unit change in x_k changes the log odds ratio by b